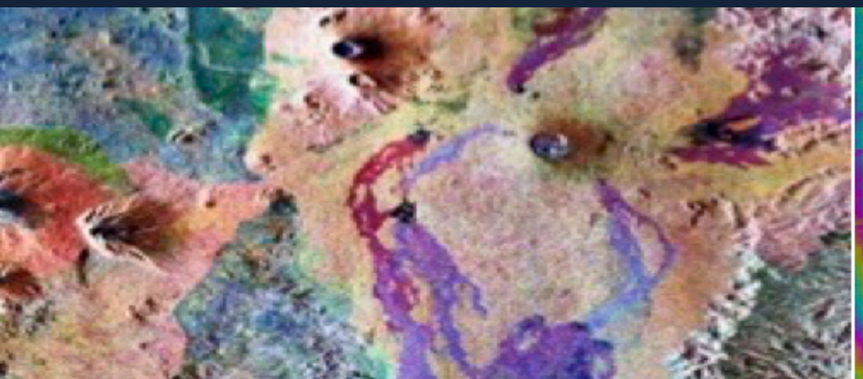
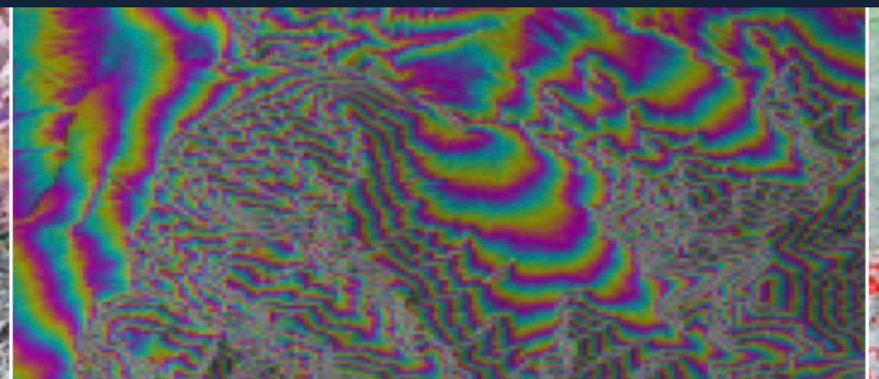
ENVIRONMENTAL GEOHAZARDS ENGINEERING · PRACTICAL

InSAR Processing with ESA SNAP

Mapping volcano deformation with Sentinel-1

AKSHAY RAJ MANOCHA







GOAL OF THIS PRACTICAL

Turn two Sentinel-1 radar images into a map showing how many centimetres a volcano’s ground moved.

Thirteen steps in SNAP, one example volcano, and the checks that tell you whether to trust the result.

THIS PRACTICAL

Outline

01 **Before you start**
Software, data, and choosing a good pair of images

02 **Processing in SNAP**
Thirteen steps from raw images to a map of ground motion

03 **Results and interpretation**
Reading the volcano deformation and estimating the magma source

04 **Real example: Kīlauea, Hawaii, 2018**
Real Sentinel-1 interferograms of the 2018 eruption and earthquake

05 **Tips and next steps**
Automating with graphs, fixing common problems, and where to go next

01

PART 1

Before you start

Software, data, and choosing a good pair of images

BEFORE YOU START

What you need

ESA SNAP

Free software for Windows, macOS and Linux, with the Sentinel-1 (Microwave) Toolbox. Download from step.esa.int.

SNAPHU

Free phase-unwrapping program by Chen and Zebker (Stanford). SNAP prepares its input and reads its output.

A capable computer

16 GB of memory recommended (8 GB minimum) and about 30 GB of free disk space.

A free data account

NASA Earthdata login (for ASF Vertex) or a Copernicus Data Space account, to download Sentinel-1 images.

An internet connection

SNAP downloads the precise orbit files and the elevation model (DEM) automatically while processing.

Time

Allow one to two hours for your first pair. Once you have a saved graph, the next pair is much quicker.

BEFORE YOU START

The workflow at a glance

01
Open and check

02
TOPS Split

03
Apply orbit

04
Back-geocoding

05
Interferogram

06
Deburst

07
Remove topography

08
Multilook

09
Goldstein filter

10
Unwrap (SNAPHU)

11
Phase to displacement

12
Terrain correction

13
Export and map

Dark steps are the milestones: the first interferogram (5), the unwrapped phase (10) and the map in real-world coordinates (12).

BEFORE YOU START

Our example and choosing a pair

Example volcano (simulated)

- A 2,650 m volcano with the sea about 13 km to the east
- A magma chamber 3 km deep gains 6 million m³ between two passes
- Sentinel-1 descending pass, 12 days apart, perpendicular baseline about 100 m
- Realistic atmospheric noise (about 0.6 cm) and speckle

Every figure in Parts 2 and 3 comes from this simulation, so you can compare your answer with the truth.

How to choose a good pair

- Same track (relative orbit) and same pass direction
- One image before and one after the change you want to see
- Short time gap (6–24 days) to keep coherence
- Small perpendicular baseline (ideally under about 150 m)
- No snow cover in either image
- Polarisation VV, mode IW, product type SLC

BEFORE YOU START

Downloading the two images

1. Go to **search.asf.alaska.edu** (ASF Vertex) and log in with your free NASA Earthdata account.
2. Draw a box around the volcano and choose the dataset **Sentinel-1**.
3. Filters: beam mode **IW**, file type **L1 Single Look Complex (SLC)**, and your date range.
4. Note a scene's **Path** (track) and **Frame**; keep only scenes with the same path and direction.
5. Open the **Baseline** tool on your reference scene to compare dates and perpendicular baselines of partners.
6. Add both scenes to the download queue (4–8 GB each) and download.

Alternative

Copernicus Browser at dataspace.copernicus.eu offers the same Sentinel-1 SLC data.

Tip

Keep the downloads as .zip files: SNAP opens them directly, so there is no need to unzip.

02

PART 2

Processing in SNAP

Thirteen steps from raw images to a map of ground motion

STEP 1 OF 13

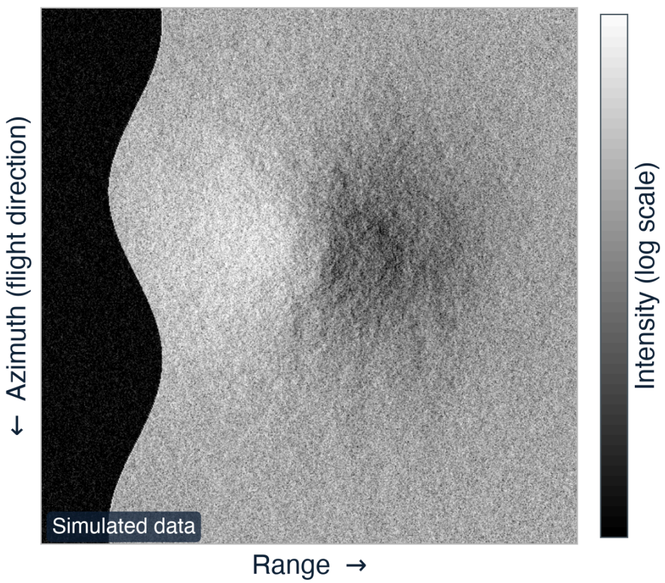
Open the two images and check them

`File › Open Product`

Setting	Value
Files	Both .zip files
Mission and mode	Sentinel-1, IW, SLC
Relative orbit	The same for both images
Pass	Both ascending or both descending

> Read **REL_ORBIT** and **PASS** in Metadata › Abstracted_Metadata.

> The earlier image is the **reference**; the later one is the **secondary**.



Intensity of the reference image, in radar geometry (mirrored). Simulated example data.

STEP 2 OF 13

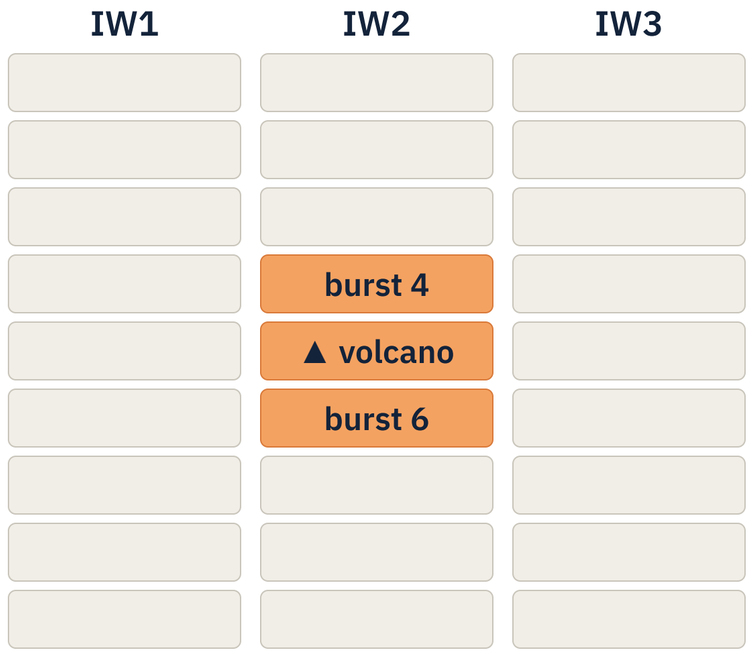
Split: keep only the bursts you need

`Radar › Sentinel-1 TOPS › S-1 TOPS Split`

Setting	Value
Subswath	The one over the volcano (IW1, IW2 or IW3)
Polarisation	VV
Bursts	The 2–4 bursts covering the volcano
Second image	Exactly the same subswath and bursts

> Processing only what you need turns hours into minutes.

> Bursts are about 20 km long in the flight direction; take one extra on each side of the volcano.



One IW image: 3 subswaths of about 9 bursts. Keep only the bursts over your target.

STEP 3 OF 13

Apply precise orbits

`Radar › Apply Orbit File`

Setting	Value
Orbit state vectors	Sentinel Precise (auto download)
Polynomial degree	3 (default)
If not found	Tick “Do not fail if new orbit file is not found”
Run on	Both split products

> Better orbits mean fewer false fringes and ramps.

> Recent images may only have restituted orbits; reprocess later for the best result.

≈ 5 cm

Accuracy of the precise orbits

≈ 20 days

Delay before precise orbits are available; restituted orbits (≈ 10 cm) come within hours

STEP 4 OF 13

Coregister the images (Back-Geocoding)

`Radar › Coregistration › S1 TOPS Coregistration › S1 Back Geocoding`

Setting	Value
Inputs	Both orbit-corrected products, reference first
DEM	SRTM 1Sec HGT or Copernicus 30 m (Auto Download)
Resampling	Defaults (bicubic DEM, bisinc image)
Option	Tick “Mask out areas without elevation”

> Aligns the secondary image to the reference, pixel by pixel, using orbits and the DEM.

> More than one burst? Next run **S1 Enhanced Spectral Diversity** (same menu, defaults).

Why so precise?

Interferometry compares phase, so the two images must line up to a tiny fraction of a pixel: about 1/1000 of a pixel in the flight direction for TOPS.

Check

The output stack holds i, q and intensity bands for both dates. Flick between the two intensities: features must not move.

STEP 5 OF 13

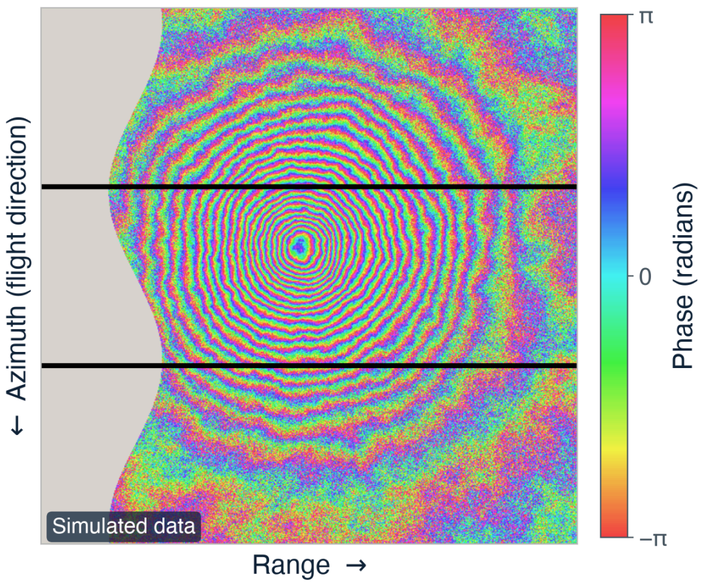
Form the interferogram

`Radar › Interferometric › Products › Interferogram Formation`

Setting	Value
Subtract flat-earth phase	Yes (defaults)
Include coherence	Yes
Coherence window	10 × 3 (range × azimuth)
Output bands	Phase, coherence, i and q

> Each pixel: reference × complex conjugate of secondary.

> The dense rings follow the volcano’s contours: about 18 fringes of **topography**, not deformation.



Raw interferogram: topographic fringes and burst gaps. Simulated example data.

STEP 6 OF 13

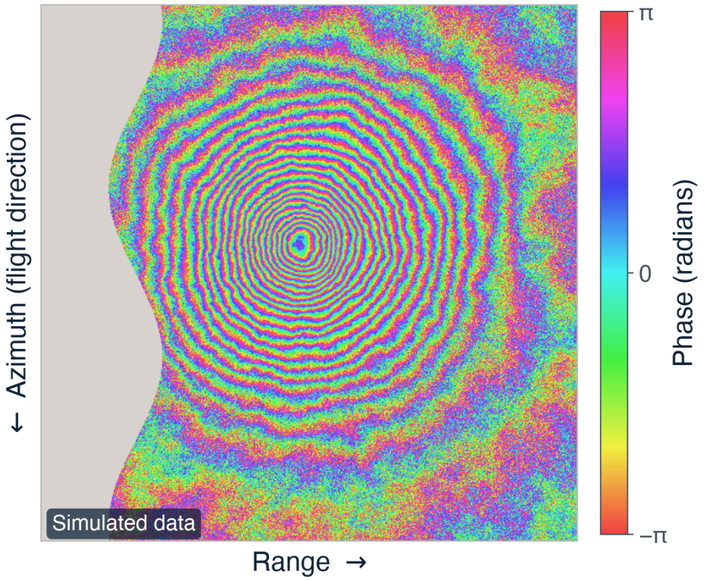
Deburst

`Radar › Sentinel-1 TOPS › S-1 TOPS Deburst`

Setting	Value
Polarisation	VV
Bands	All (phase, coherence, i, q)

> Joins the bursts into one seamless image and removes the black lines between them.

> Several subswaths? Merge them now with S-1 TOPS Merge.



After deburst: one continuous interferogram. Simulated example data.

STEP 7 OF 13

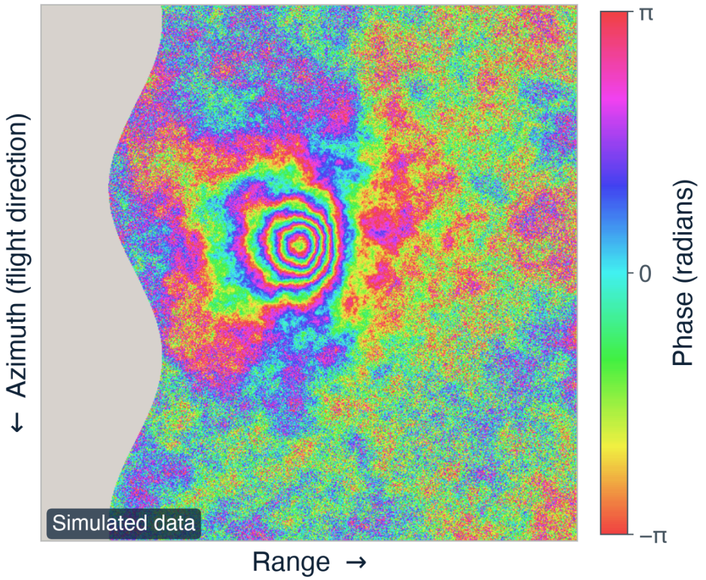
Remove the topographic phase

`Radar › Interferometric › Products › Topographic Phase Removal`

Setting	Value
DEM	Same as step 4 (Auto Download)
Optional outputs	Elevation band (useful later for checking atmosphere)

> SNAP simulates the phase the topography would produce and subtracts it.

> What remains: **deformation + atmosphere + noise**. Here about 5 rings around the summit: the volcano is inflating.



After removing topography: deformation fringes. Simulated example data.

STEP 8 OF 13

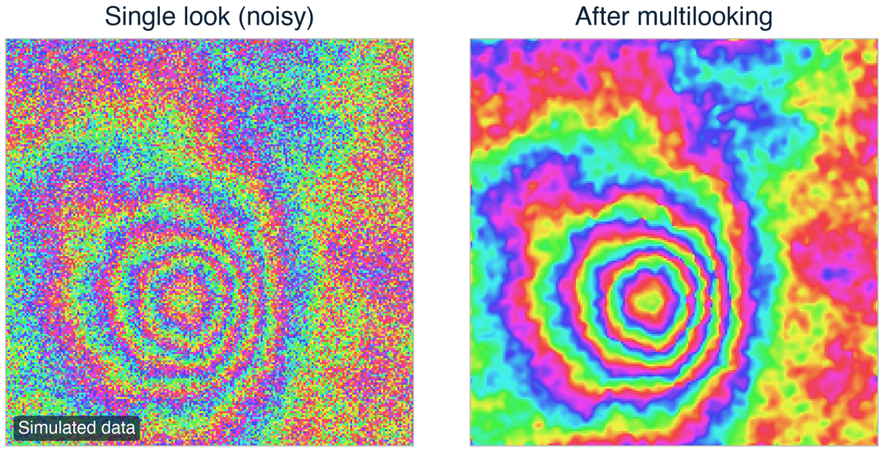
Multilook

`Radar › SAR Utilities › Multilooking`

Setting	Value
Looks	Independent looks
Range × azimuth looks	4 × 1 (or 8 × 2 for noisy areas)
Output pixel	About 15 m × 14 m

> Averages neighbouring pixels: less noise, smaller files and square pixels.

> More looks = cleaner phase but less detail.



Summit area before and after multilooking. Simulated example data.

STEP 9 OF 13

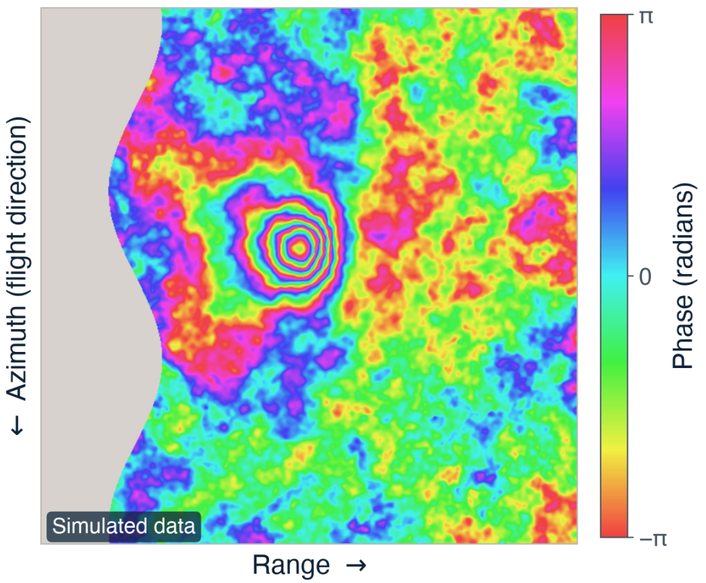
Filter the phase (Goldstein)

`Radar › Interferometric › Filtering › Goldstein Phase Filtering`

Setting	Value
Adaptive filter exponent	1.0
FFT size	64
Window size	3
Coherence mask	Optional, threshold 0.2

> Smooths noise while keeping fringe edges sharp: easier to read and to unwrap.

> Strong filtering can make noise look like fringes: always compare with the unfiltered phase.



After Goldstein filtering. Simulated example data.

STEP 10 OF 13

Unwrap the phase with SNAPHU

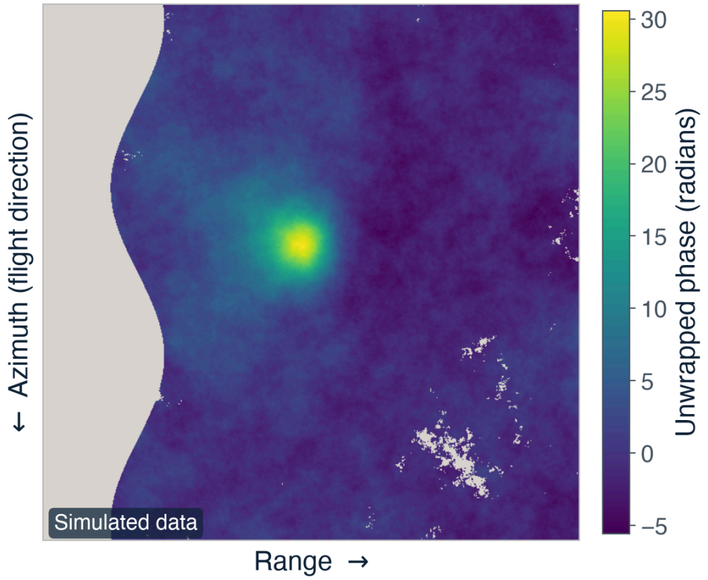
`Radar › Interferometric › Unwrapping › Snaphu Export → run snaphu → Snaphu Import`

Setting	Value
Statistical-cost mode	DEFO (deformation)
Initial method	MCF
Tiles	1 × 1 for small areas (more, with overlap, for large ones)
Target folder	A short path without spaces

In a terminal, inside the export folder, run the command printed at the top of **snaphu.conf**, for example:

```
snaphu -f snaphu.conf Phase_ifg_VV.snaphu.img 2560
```

> Then **Snaphu Import**: choose the filtered product and the UnwPhase .hdr file. Recent SNAP versions can also run SNAPHU for you.



Unwrapped phase: continuous, no more 2π jumps. Simulated example data.

STEP 11 OF 13

Convert phase to displacement

`Radar › Interferometric › Products › Phase to Displacement`

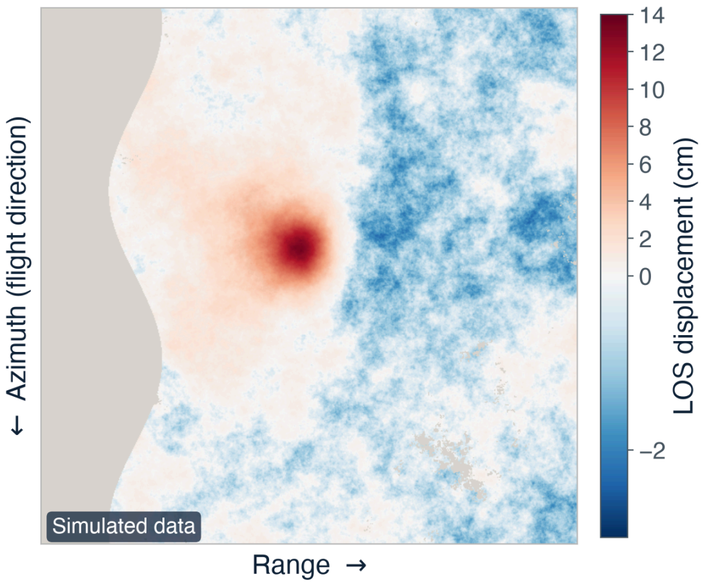
$$d = -\lambda \cdot \varphi / 4\pi$$

λ = 5.55 cm for Sentinel-1, so one fringe (2π) = 2.8 cm along the line of sight.

The output is line-of-sight (LOS) displacement in metres.

Pick a **stable reference area** far from the volcano and subtract its mean (Raster › Band Maths), so zero really means no motion.

Check the sign: read the band description and compare with a known area or GNSS before interpreting.



LOS displacement, still in radar geometry. Simulated example data.

STEP 12 OF 13

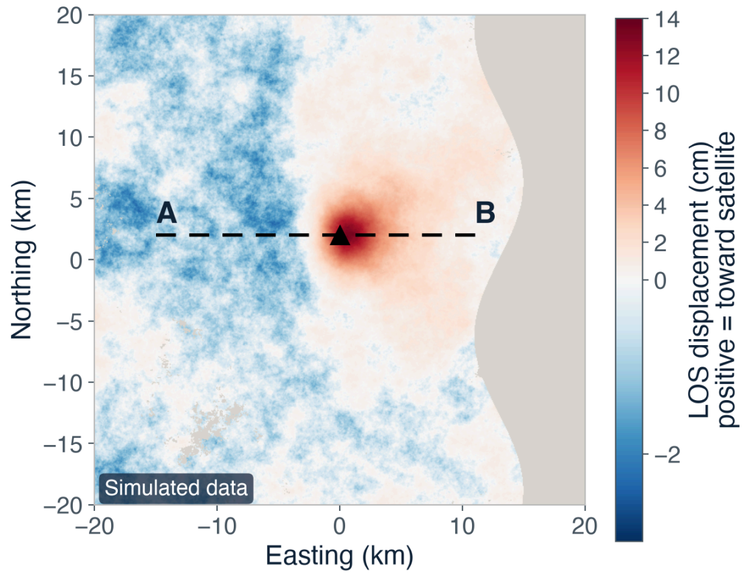
Terrain correction (geocoding)

`Radar › Geometric › Terrain Correction › Range-Doppler Terrain Correction`

Setting	Value
Source bands	Displacement and coherence
DEM	SRTM 1Sec or Copernicus 30 m (Auto Download)
Pixel spacing	15 m (or 30 m)
Map projection	WGS84 or UTM

> Moves every pixel from radar geometry (mirrored and slanted) to map coordinates, north up.

> Now the result lines up with maps, roads, faults and towns in GIS.



Geocoded displacement with profile A–B. Simulated example data.

STEP 13 OF 13

Export and make a map

`File › Export › GeoTIFF · right-click image › Export View as Google Earth KMZ`

Setting	Value
Colour	Colour Manipulation: blue–red palette, centred on zero
Mask	Band Maths: hide coherence below 0.3
Formats	GeoTIFF for QGIS; KMZ for Google Earth

> In QGIS, add a hillshade, towns and faults, and draw a profile (Profile Tool plugin).

> Always label the map so others can read it without you.

A good InSAR map shows

- The two dates (reference → secondary)
- Satellite, track and pass direction
- An arrow for the line of sight
- A colour bar in cm, with the sign explained
- Low-coherence areas masked
- A scale bar and north arrow

03

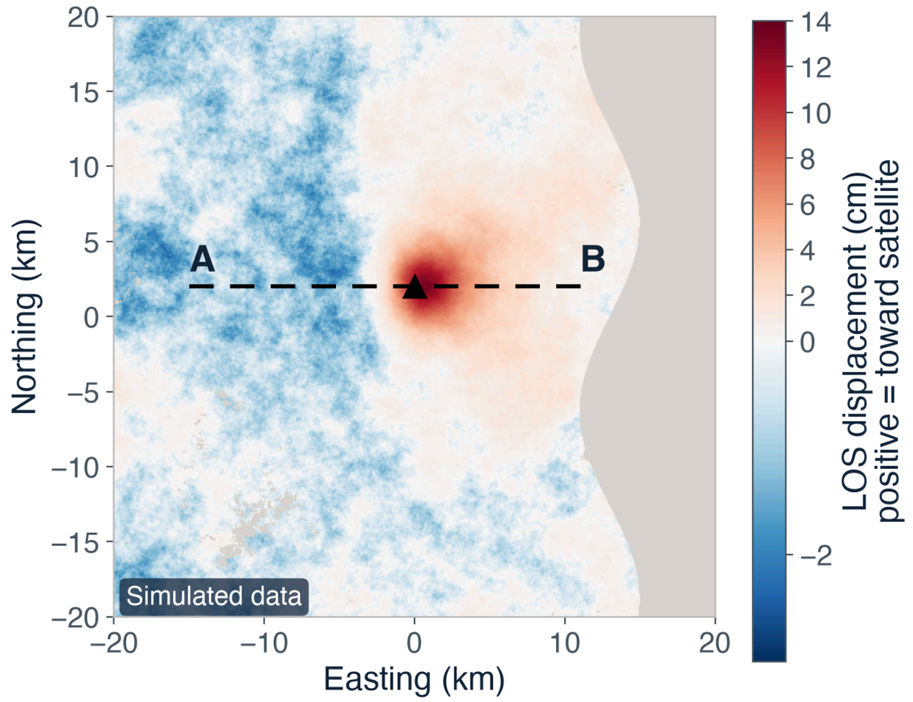
PART 3

Results and interpretation

Reading the volcano deformation and estimating the magma source

RESULTS

The volcano is inflating



Final map: line-of-sight displacement. Simulated example data.

13.5 cm
Peak motion toward the satellite, near the summit

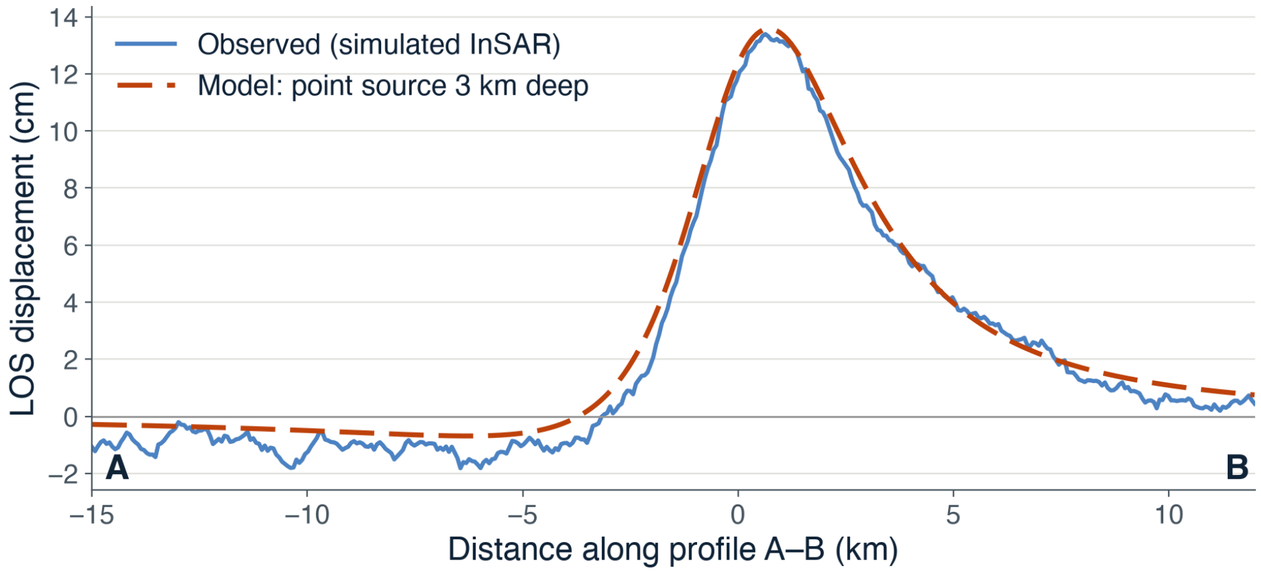
≈ 5
Fringes between the edge of the bulge and its centre

≈ 14 km
Width of the area moving more than 1 cm

≈ 0.6 cm
Background noise, mostly atmosphere

RESULTS

A profile across the volcano



Profile A–B through the summit. Simulated example data.

Observed vs model

The data and a point source 3 km deep agree to within about 1 cm; the wiggles are atmosphere.

Why is the peak off-centre?

The satellite looks from the east, so it also sees the ground moving east: the bulge appears shifted toward the satellite.

INTERPRETATION

How deep is the magma, and how much?

A simple model treats the magma chamber as an inflating point (the **Mogi model**). Two rules of thumb then give its depth and volume from the map:

1 Depth

$$d \approx 1.3 \times r_{1/2}$$

$r_{1/2}$ = distance from the centre where uplift falls to half its peak. From the profile: $r_{1/2} \approx 2.3$ km, so $d \approx 3.0$ km.

2 Uplift

$$u \approx \mathrm{LOS} / \cos\theta$$

Assuming mostly vertical motion: 13.6 cm / cos 39° ≈ 17.5 cm.

3 Volume

$$\Delta V \approx \pi\, d^2\, u / (1 - \nu)$$

With ν = 0.25: π × (3,000 m)² × 0.175 m / 0.75 ≈ 6.6 million m³.

Truth in the simulation: depth 3.0 km, volume change 6.0 million m³. The estimate is close, and slightly high because the line of sight also includes horizontal motion.

INTERPRETATION

Is the signal real?

Repeat with another pair

Real deformation appears in independent pairs over the same period; the atmosphere changes from date to date.

Compare with elevation

On tall volcanoes, the atmosphere can make fringes that follow the contour lines. Use the elevation band from step 7 to check.

Use both passes

Process an ascending pair too. Deformation should make sense from both directions; then you can separate vertical and east–west motion.

Check against other data

GNSS stations, tiltmeters, observatory reports, and ready-made interferograms from the COMET-LiCSAR volcano portal.

04

PART 4

Real example: Kīlauea, Hawaii, 2018

Real Sentinel-1 interferograms of the 2018 eruption and earthquake

REAL EXAMPLE

What happened at Kīlauea in 2018

30 April
The crater floor of Puʻu ʻŌʻō collapses; magma moves down the East Rift Zone

3 May
Lava erupts from fissures inside Leilani Estates, a residential area

4 May
A magnitude 6.9 earthquake strikes the south flank, which slips toward the sea

May–August
The summit caldera collapses, step by step

September
Lava stops erupting after four months

62
Summit collapse events

500+ m
Deepening of the summit crater

35.5 km²
Land covered by new lava

700+
Homes and other structures destroyed

Facts: USGS Hawaiian Volcano Observatory. Sentinel-1 imaged the island every 6 days throughout.

REAL EXAMPLE

The data: two real Sentinel-1 pairs

Setting	Value
Satellites	Sentinel-1A and 1B, C-band (5.55 cm)
Track	Relative orbit 124, ascending (looking east)
Pair 1	20 April → 2 May 2018 (12 days): magma intrusion before the eruption
Pair 2	2 May → 8 May 2018 (6 days): eruption onset and the M6.9 earthquake
Processing	COMET-LiCSAR, frame 124A_06996_091406: topography removed, filtered, unwrapped with SNAPHU

These are the same products you make in SNAP steps 5–12, computed automatically by COMET for every new Sentinel-1 pair.

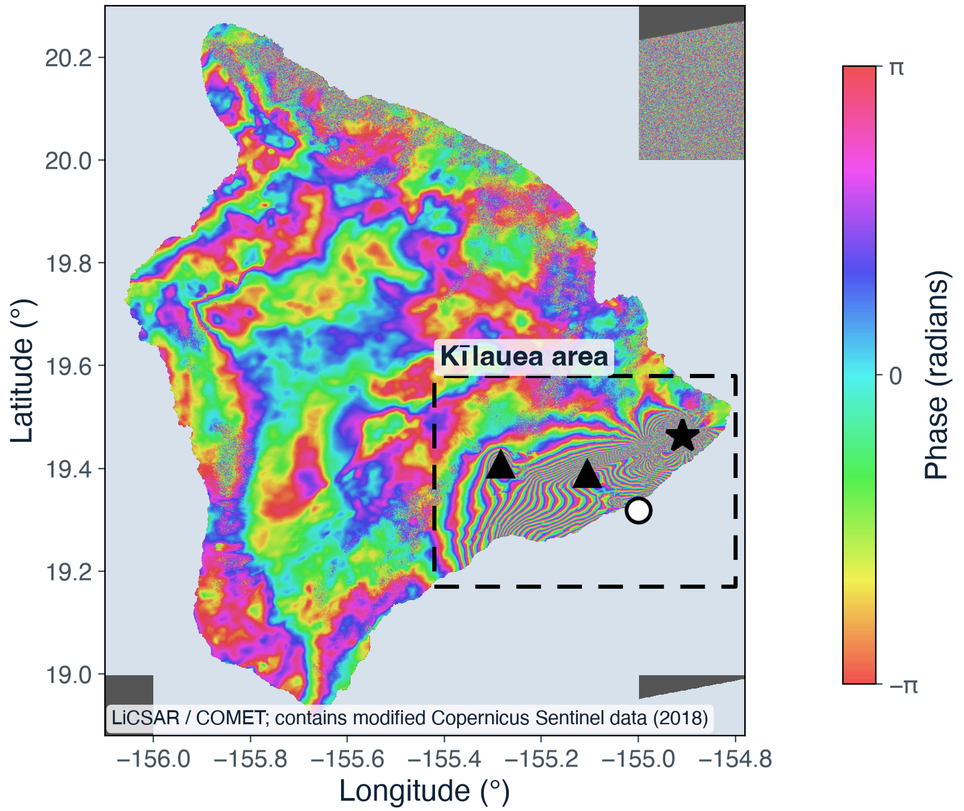
Data credit

LiCSAR contains modified Copernicus Sentinel data 2018 analysed by the Centre for the Observation and Modelling of Earthquakes, Volcanoes and Tectonics (COMET). LiCSAR uses JASMIN, the UK's collaborative data analysis environment (jasmin.ac.uk).

Lazecky and Maghsoudi (2022), LiCSAR interferometry products, NERC EDS CEDA. Open Government Licence v3.

REAL EXAMPLE

Deformation or atmosphere?



Whole island, 2 → 8 May 2018 (wrapped phase).

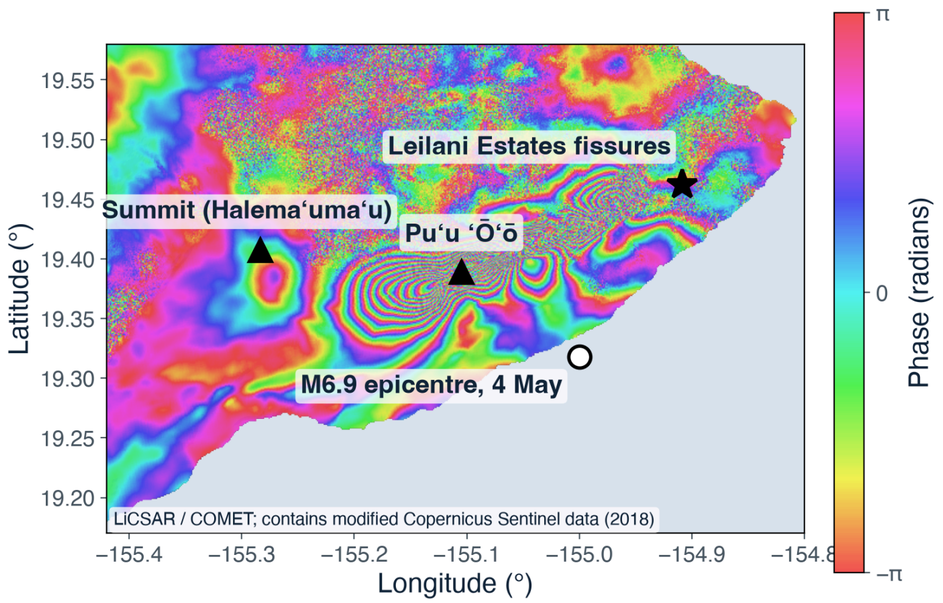
Tight, dense fringes on Kīlauea's south-east flank: **real deformation** from the intrusion and the earthquake.

Broad, blotchy fringes over Mauna Loa and Mauna Kea (over 4,000 m high): mostly **atmosphere**, because water vapour changes with height.

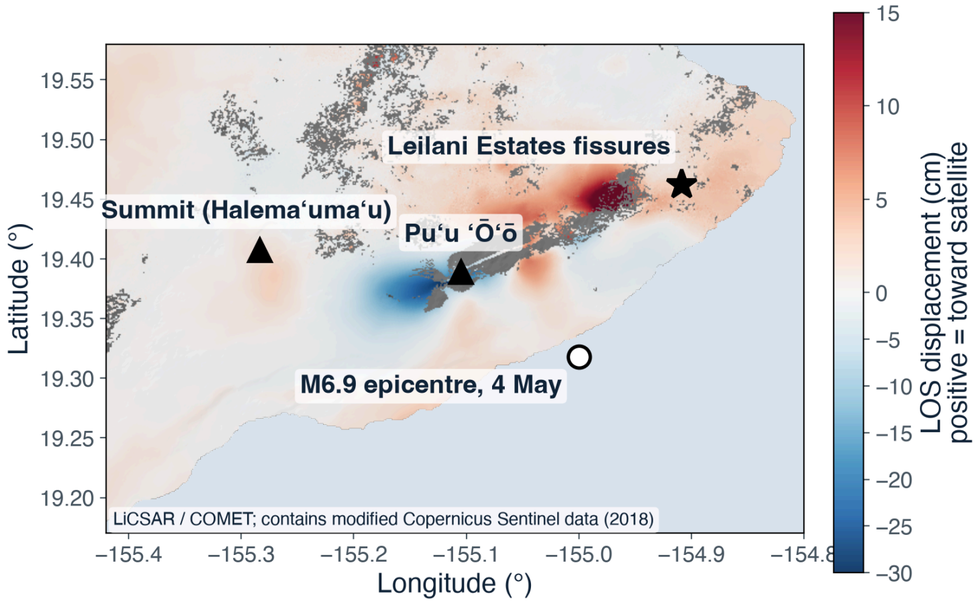
Test: atmospheric fringes follow the topography and change from pair to pair; deformation repeats in every pair that spans the event.

REAL EXAMPLE

Before the eruption: magma on the move



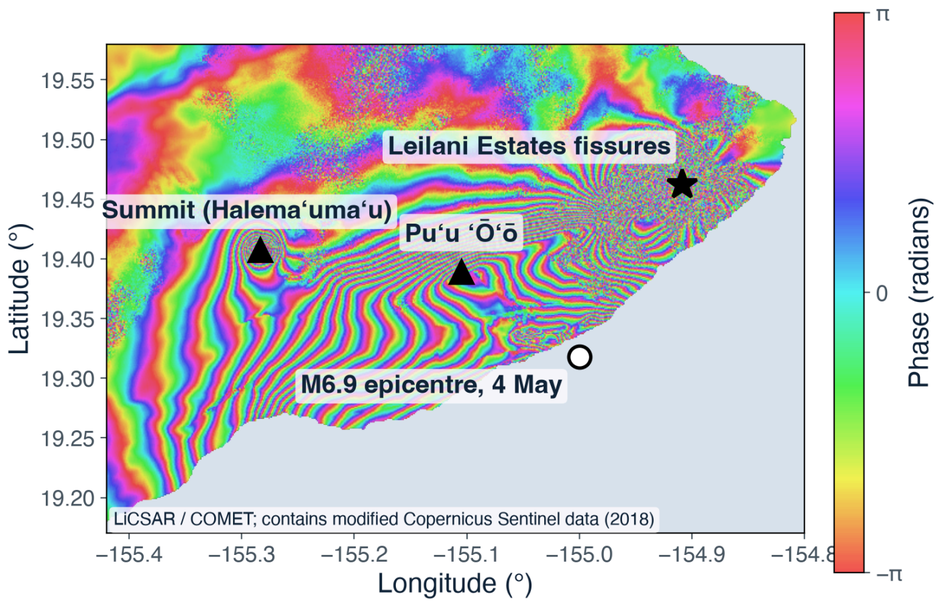
20 April → 2 May: fringes around Pu‘u ‘Ō‘ō and along the lower rift.



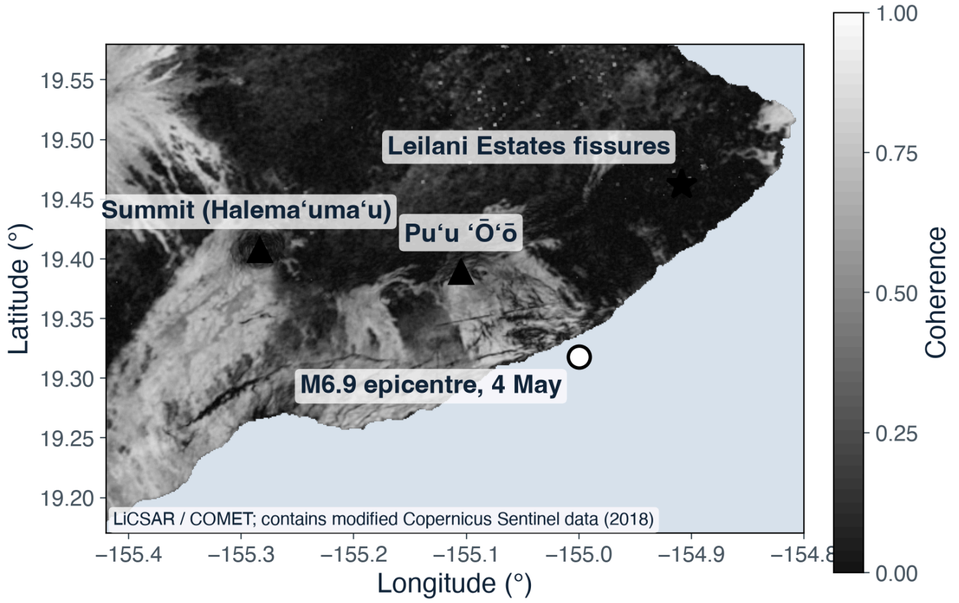
Pu‘u ‘Ō‘ō drops (blue) as its crater collapses; the lower rift rises (red) as magma intrudes.

REAL EXAMPLE

Eruption and earthquake: phase and coherence



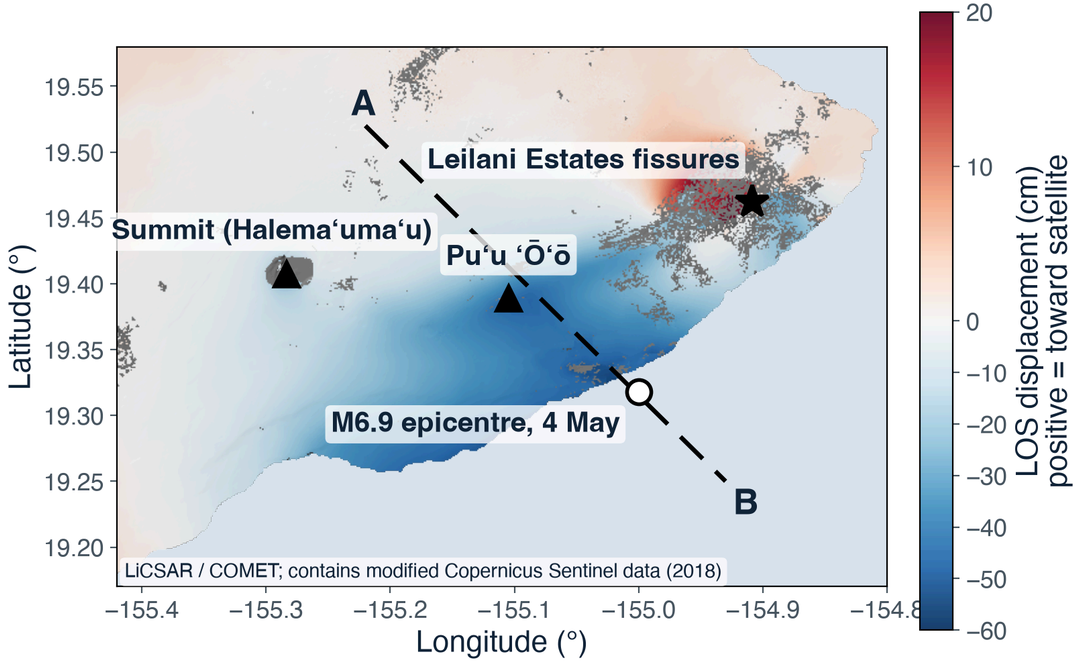
2 → 8 May: about 20 fringes across the south flank.



Coherence: low on the wet, forested east and on new lava; high on the bare lava fields to the south-west.

REAL EXAMPLE

The displacement map: 2 → 8 May 2018



LOS displacement; dashed line = profile A–B.

The south flank moved up to about **55 cm away** from the satellite: it slid seaward and down in the M6.9 earthquake.

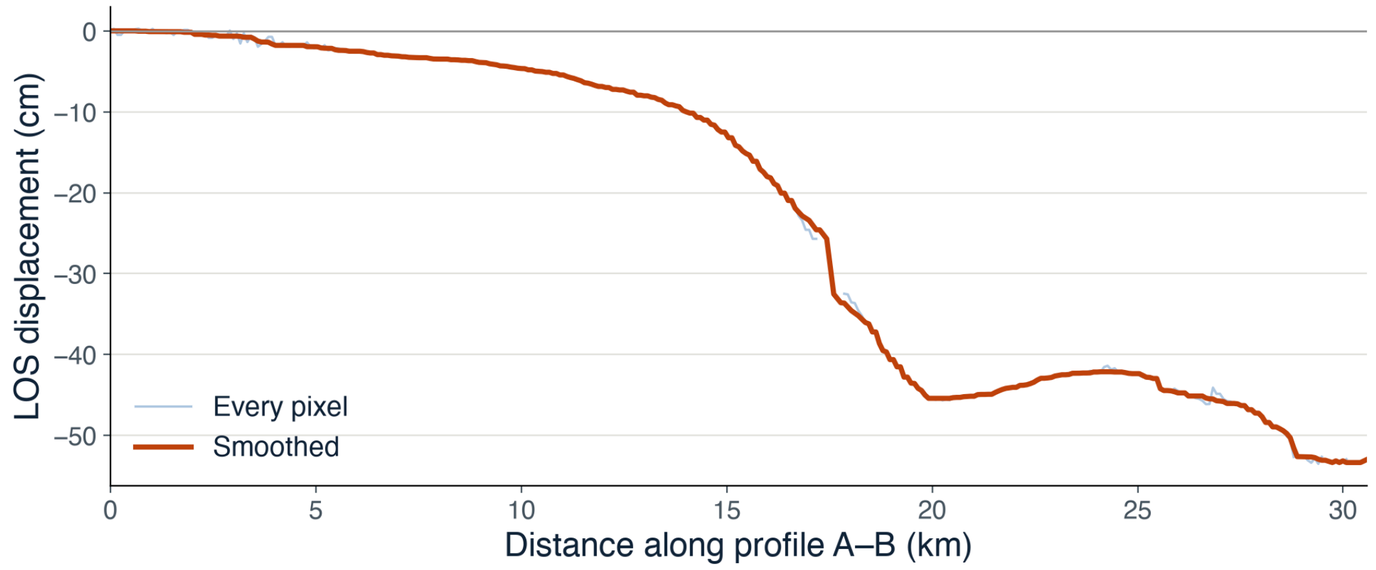
Around the Leilani fissures the ground moved **toward** the satellite, up to about 20 cm: the rift opening as the dike widened.

Grey gaps are low-coherence areas, left blank: new lava and dense forest.

Positive = toward the satellite (ascending pass, looking east).

REAL EXAMPLE

Profile A–B across the south flank



Profile A–B, 2 → 8 May 2018 (line on the previous map).

Check your unwrapping

The profile drops by about 53 cm, so the wrapped image should show about 19 fringes along A–B (53 ÷ 2.8). Count them two slides back.

Real data is messier

Atmosphere, gaps from low coherence, and several processes on top of each other: intrusion, earthquake and summit deflation.

PRACTICE

Your turn: reproduce Kīlauea in SNAP

Kīlauea, 2 May → 8 May 2018

- ASF Vertex: Sentinel-1, IW, L1 SLC, path 124, ascending
- Reference 2 May 2018, secondary 8 May 2018
- TOPS Split: the subswath and 3–4 bursts covering the east rift and south flank
- Run the 13 steps; use 8 × 2 looks for the dense fringes
- Compare your map with the COMET-LiCSAR result in this lecture

Also try: Etna, Italy

December 2018 eruption and the magnitude 4.9 Fleri earthquake: one image before 24 December, one after.

Campi Flegrei, Italy

Slow, ongoing uplift of the caldera since 2005: ideal for time series.

Sierra Negra, Galápagos

Uplift before and during the June 2018 eruption.

05

PART 5

Tips and next steps

Automating with graphs, fixing common problems, and where to go next

TIPS

Automate with the Graph Builder

Chain steps 1–9 into one graph and run it in one go:

`Read ×2` `TOPSAR-Split ×2` `Apply-Orbit-File ×2`
`Back-Geocoding` `Enhanced-Spectral-Diversity` `Interferogram`
`TOPSAR-Deburst` `TopoPhaseRemoval` `Multilook`
`GoldsteinPhaseFiltering` `Write`

A second graph can do Snaphu import, Phase to Displacement and Terrain Correction.

Build

Tools › GraphBuilder: add operators, connect them, set the parameters, then save the graph as an XML file.

Repeat

Tools › Batch Processing runs the same graph on many pairs.

Command line

```
gpt my_insar_graph.xml
```

gpt is SNAP’s Graph Processing Tool.

TIPS

Troubleshooting

Problem	Likely cause	Fix
Back-Geocoding fails or the images don’t overlap	Different tracks, or different bursts kept	Check REL_ORBIT; redo TOPS Split with matching bursts
DEM or orbit download error	No internet, or server busy	Retry later or choose another DEM (Copernicus 30 m)
SNAP runs out of memory or is very slow	Too many bursts for the RAM	Split fewer bursts; raise SNAP’s memory in Tools › Options › Performance
Interferogram is pure noise	Low coherence: long gap, vegetation or snow	Choose a shorter pair; multilook more; try L-band (NISAR)
Phase jumps at burst edges	ESD skipped with several bursts	Run Enhanced Spectral Diversity after Back-Geocoding
Unwrapped result has sudden steps	Unwrapping errors in noisy areas	Mask coherence below 0.3, filter more, unwrap again

TIPS

Good-practice checklist

While processing

- Same track and pass direction
- Short time gap; no snow
- Precise orbits applied
- Same bursts in both images
- ESD run if more than one burst

Before you interpret

- No fringes left following the contours
- Low-coherence areas masked
- Reference area chosen and set to zero
- Sign convention checked
- Map labelled: dates, pass, LOS, units

RECAP

Key steps

01 Choose a good pair: same track, short time gap, small baseline, no snow.

02 Split, orbit, coregister, then form the interferogram and remove topography.

03 Multilook and filter, then unwrap with SNAPHU and convert to centimetres.

04 Geocode, mask low coherence, set a reference area and label the map.

05 Check the signal is real before interpreting: other pairs, both passes, ground data.

NEXT STEPS

From one interferogram to a time series

StaMPS

Persistent-scatterer time series. SNAP can export a stack of interferograms in StaMPS format.

MintPy

Small-baseline time series in Python, from ISCE2 or ASF HyP3 interferogram stacks.

LiCSBAS

Time series from ready-made COMET-LiCSAR interferograms: no SLC processing at all.

Thank you. Questions?

Software: ESA SNAP (step.esa.int) and SNAPHU by C. W. Chen and H. A. Zebker (Stanford University). Sentinel-1 data: EU Copernicus programme / ESA. Kīlauea 2018 products: COMET-LiCSAR (Lazecky and Maghsoudi, 2022), which contains modified Copernicus Sentinel data 2018 analysed by COMET and uses JASMIN; event facts from the USGS Hawaiian Volcano Observatory. The volcano example in Parts 2 and 3 is simulated (a Mogi point source beneath a synthetic volcano, with Sentinel-1 wavelength and geometry) so that results can be checked against a known answer.